

Budget Buddy

AI-Powered Personal Finance Manager

Data Science Capstone Project

Phase 1: Problem Identification, Domain Research & Use-Case Finalization

GROUP - 13

April 2026



1. Objective

To develop an intelligent system that helps students and users track expenses, understand their spending habits, and improve their saving behaviors through personalized insights.

2. Domain

- Finance (FinTech) — AI-driven personal finance management
- EdTech — Financial literacy tools for college students and young professionals

3. Research Methods

Blogs / Articles

1. Building an AI-Driven Personal Finance App
2. How to Build a Personal Finance App like Mint: Ultimate Guide for 2022

Research Papers

3. An AI-Powered Personal Finance Assistance Enhancing Financial Literacy and Management

4. AI-Based Voice Assistant Systems

4. Research Focus Areas

- Spending Pattern Analysis
- Expense Prediction (ML focus)
- Budget Monitoring & Alerts
- User Behaviour Analysis
- Voice Expense Input

5. Data Pain Points

5. Poor saving habits
6. Lack of financial planning
7. Existing apps are too manual
8. Lack of personalized insights
9. Low user engagement
10. Lack of spending pattern awareness
11. Overspending without alerts

6. Project Scope

- Description Analysis — Track and summarize user expenses
- Prediction Analysis — Predict next month's expenditures
- Predictive Analysis — Suggest basic saving recommendations

7. Existing Solution Gaps

Existing Apps

- Walnut
- Mint
- Money Manager

Identified Gaps

- No expense prediction
- Limited personalization
- No behaviour analysis

- Poor engagement
- No smart budget alerts

8. Use-Case Summary

Analytical Questions

12. What is the user's monthly spending?
13. Which category has the highest spending?
14. Is the user overspending?
15. What is next month's predicted expense?
16. How much can the user save?

Inputs	Outputs
• Expense Amount • Category • Date • Budget Limit • Income	• Spending Summary • Category-wise Analysis • Future Expense Prediction • Budget Alerts • Saving Suggestions

9. Target Users & Stakeholders

Target Users

- Job Professionals
- College Students
- Over-Spenders
- Savers

10. Data Research

The following datasets and sources will be leveraged for model training and validation:

- Kaggle — Personal Finance Datasets (expense tracking, income, budget records)
- Public APIs — Banking/transaction data APIs for real-time expense input
- Synthetic Data Generation — For initial prototyping and model testing
- User Surveys — Primary data collection from target users for validation

11. Phase 1 Deliverables

- 17. Problem Statement Document (1 Page)
- 18. Domain & Data Research Summary
- 19. Use-Case Summary (this document)
- 20. Stakeholder List